Project Graphics Report

1- Game Title : OpenTrack Racer

Team Members :

1- Ahmed Ashraf 320230075 (Lead Developer) (TL)

2- Rawan Hossam 320230068 (Creative Director)

3- Ahmed Haytham 320230076 (HTML & CSS Dev)

2-Tools & Frameworks Used :

1- HTML5 Canvas

- Used for rendering the game graphics.
- Allows fast 2D drawing and real-time updates.

2- JavaScript (Vanilla JS)

- Core game logic, physics, AI behavior, and rendering loop.
- No external game engine used → shows understanding of fundamentals.

3 - CSS

- Styling the UI elements (HUD, menus, settings panel).

4- Browser APIs

- **requestAnimationFrame → smooth animation loop.**
- **localStorage → saving best lap times.**
- **Keyboard Events → player controls.**

2. Game Overview & Story / Objective

Game Type: Arcade-style racing game.

Objective:

- **Control a car on a procedurally generated road.**
- **Avoid collisions with AI cars.**
- **Complete laps as fast as possible.**
- **Achieve the best lap time.**

Gameplay Features:

- **Increasing difficulty with speed.**
- **AI cars that change lanes and avoid collisions.**
- **Curves, hills, and slopes to simulate a real road.**
- **Real-time HUD showing speed and lap times.**

3.Graphics Techniques Used

A) Camera Transformations

- The game uses a pseudo-3D camera system.
- Objects are transformed from world space → camera space → screen space.

- Implemented using the project() function :

screenX = centerX + (scale * cameraX)
screenY = centerY - (scale * cameraY)

This creates depth and perspective without true 3D rendering.

B) Perspective Projection (Fake 3D)

Objects farther away appear smaller.

Controlled by:

- cameraDepth
- fieldOfView
- cameraZ

This simulates 3D using mathematical projection, not WebGL.

C) Lighting

- No real-time lighting engine.
- Simulated lighting using:
- Color shading
- Road segment alternation
- Fog fading based on distance

Util.exponentialFog(distance, density)

This creates depth and atmosphere efficiently.

D) Texture Mapping

- Sprites (cars, trees, billboards) are 2D images.
- Placed in 3D world positions and projected onto the screen.
- Scaling changes with distance to simulate depth.

E) Shaders

❌ No shaders used
Reason:

- Canvas 2D rendering does not support shaders.
- Instead, mathematical projection and scaling are used.

(This is NOT a weakness — It's a design decision.)

F) Animations

- Continuous animations using:
- requestAnimationFrame
- Delta time (dt)

- <u>Includes:</u>

- Car movement
- Steering animation
- Road scrolling
- Background parallax movement

```
update(dt)
render()
```

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4. How to Run the Game :

Steps:

- **Download or clone the project files.**

- **Open the main index.html file.**

Run it in any modern browser:

- **Chrome**
- **Firefox**
- **Edge**

Controls:

- **Arrow Up/W - Accelerate**

- **Arrow Down/S - Brake**

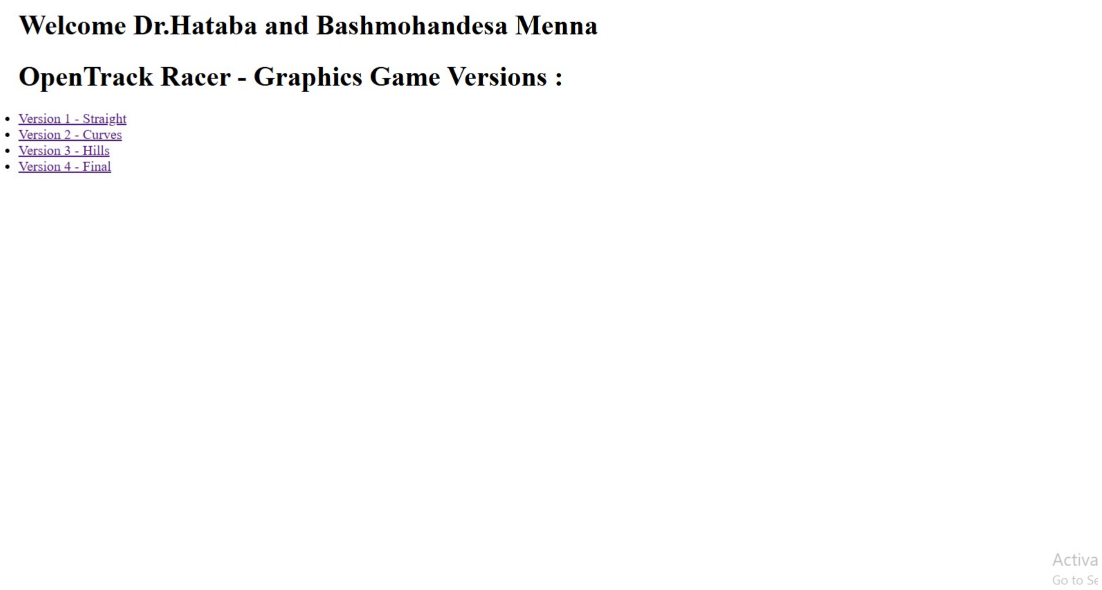
- **Left Arrow/A Right Arrow/D -- Steer left & right**

No installation required

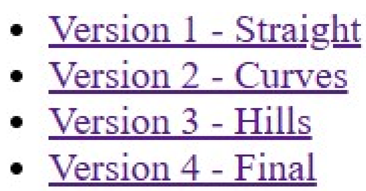
- **No server**
- **No dependencies**
- **Works offline**

Screenshots of the Game :

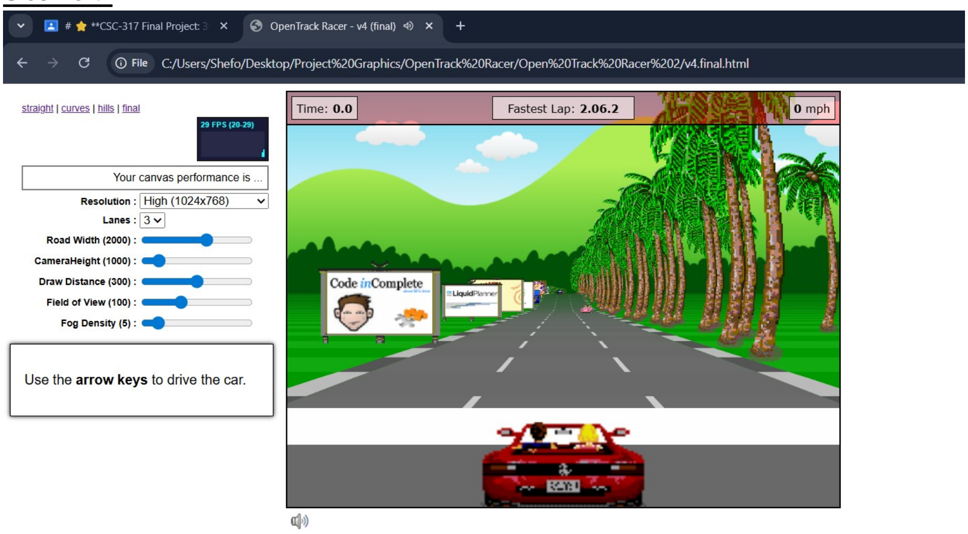
Welcome Page :



Levels / Demos : (We Choose V4 as it's the final version of the game)

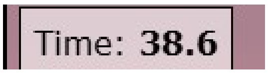


Start :

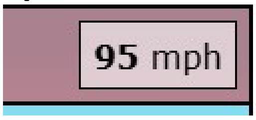


Gameplay :

Time Counter :

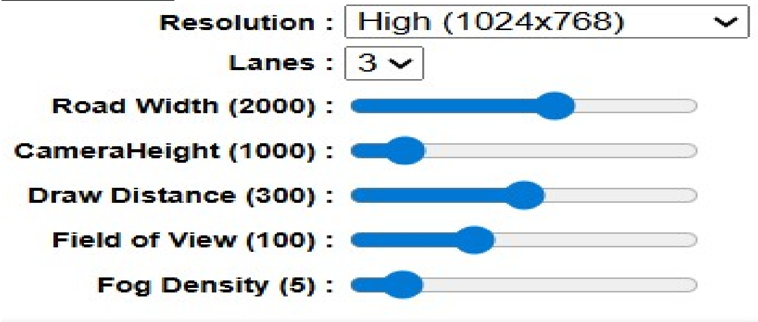


SpeedoMeter :

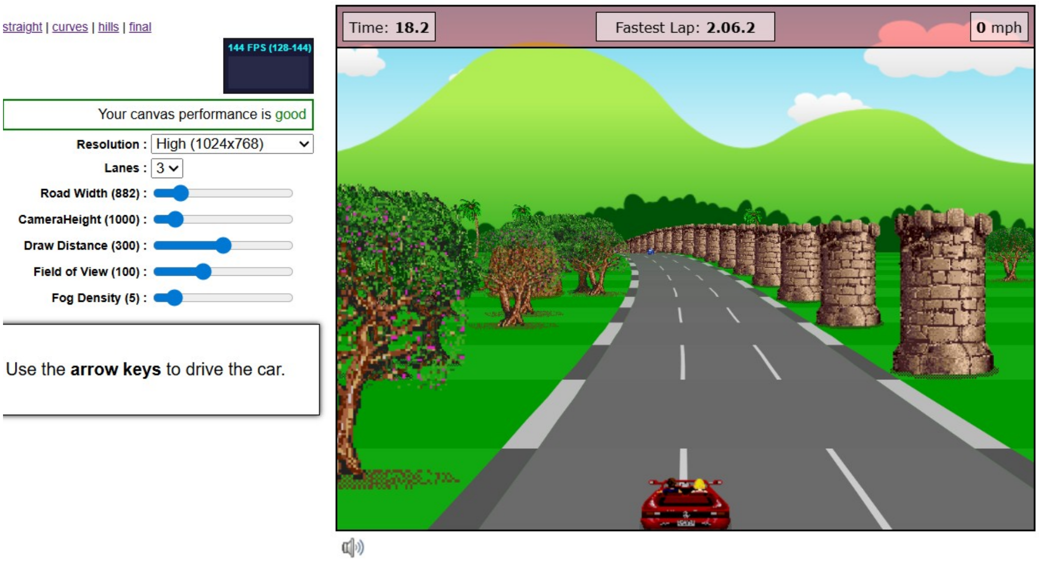


HighScore :

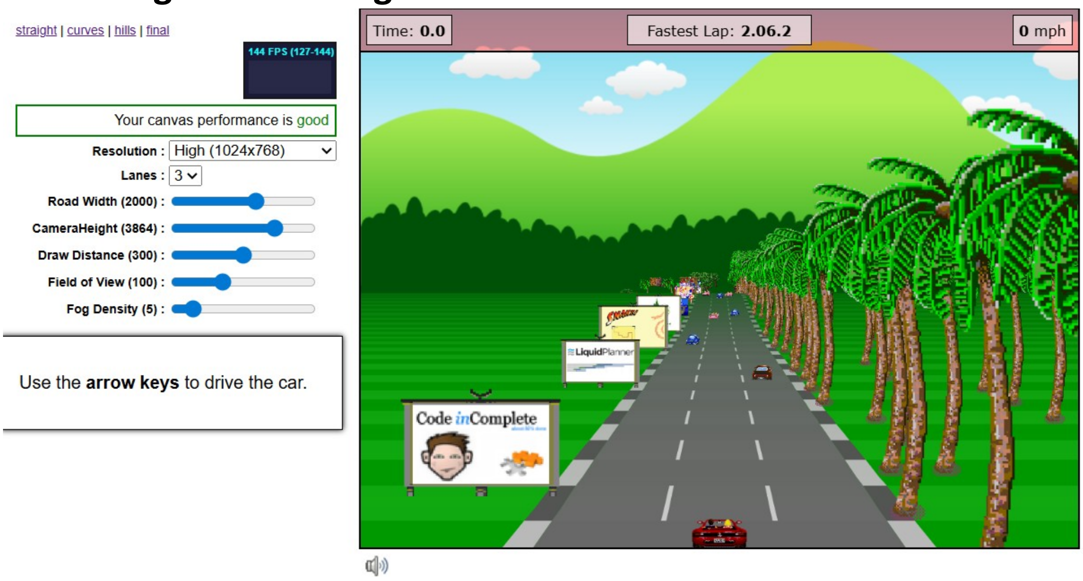
Features :



Changing Road Width :



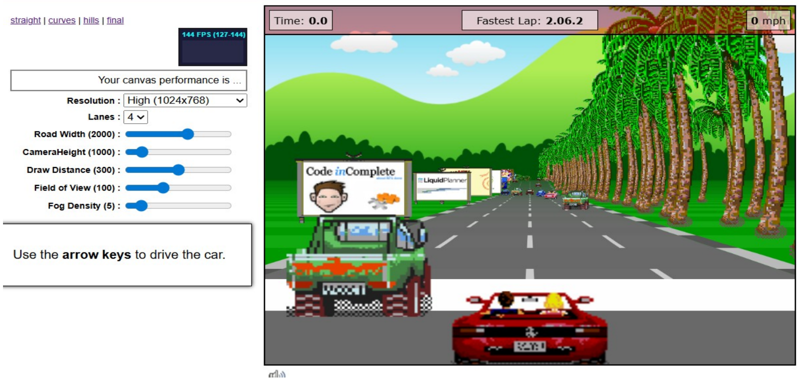
Increasing Camera Height :



Increasing draw distance :

Increasing field of view :

Increasing number of lanes :



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